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Bachelor Thesis in Physics
submitted by

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born in Stuttgart (Germany)

2026

About ...

This Bachelor Thesis has been carried out by Ansel Beucker at the
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Abstract

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Zusammenfassung

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Contents

Notation

Standard probability theory notation is adopted. In particular, the following notations are used:

- X for random variables with sample space $\mathcal{X}$
- $\mathcal{P}(\mathcal{X})$ for the set of all probability measures on the Borel σ-algebra over $\mathcal{X}$
- $P \in \mathcal{P}(\mathcal{X})$ for probability measures
- p_X for the density function of a probability measure P, index might be dropped if notation is clear
- $X \sim p$ for a random variable X whose distribution is described by p

When discussing neural networks, the symbol q or q_θ will be used for the model of a true distribution p.

1 Introduction

The current cosmological standard model predicts an epoch called the Dark Ages, during which the universe is already matter-dominated, the cosmic microwave background (CMB) has already formed and most baryonic matter consists of a gas of neutral hydrogen and helium, with some additional trace elements from big bang nucleosynthesis. This is followed by the epoch of Cosmic Dawn (CD), during which the universe's first galaxies and structures form, and a subsequent Epoch of Reionization (EoR), during which the intergalactic medium (IGM) is ionized by these new sources of radiation. The EoR is estimated to have taken place at a redshift range of roughly $z \sim 5-20$. A number of sources from multi-messenger astronomy have contributed to this current model: Scattering of CMB photons on free electrons during and after reionization can be detected once the IGM became ionized significantly enough. This places the beginning of EoR at $z > 8$ [1], though with a large uncertainty. Lyman-alpha forests in the spectra of distant quasars and galaxies are used to estimate the neutral hydrogen fraction at lower redshifts, placing the end of the EoR at around $z \sim 5-6$ (see [2]). The JWST with an observational range of up to $z \sim 15$ measures earliest galaxies is already contributing to a more detailed mapping of the neutral hydrogen fraction (e.g. [3]).

A comparatively new probe is that of the 21 cm hyperfine transition line of neutral hydrogen, which constitutes a background spectrum similar to that of the CMB. It is the only known probe with which the Dark Ages could be studied and also the one to study CD to the highest redshifts. First radio telescopes such as LOFAR are attempting to put first constraints on the EoR, though the background noise has proved to be a significant challenge (need citation). Much better results are expected from the Square Kilometer Array (SKA) Observatory, a radio telescope currently in construction which will image the 21 cm line of the IGM at a redshift range of $z \sim 6$ to $20-25$ with unprecedented accuracy. Hopes are that this signal can be used to significantly improve the current understanding of the reionization history and structure formation and possibly constrain important astrophysical and cosmological parameters.

There are multiple difficulties in interpreting the 21 cm signal. Traditional Gaussian summary statistics do not capture enough of the signal's structure, reflecting the fact that the processes being observed are highly nonlinear. This also entails that the likelihood of the 21 cm signal is intractible. Additionally, the foreground contamination for ground-based observatories such as the SKA is significant. Lastly, modern physics experiments generate an extreme amount of data, which has to be efficiently processed.

These challenges motivate the use of neural networks (NN) for the analysis of 21 cm data. Neural networks are nowadays commonly used in Bayesian inference, which attempts to reconcile prior knowledge and empirical data in order to find optimal parameters describing, for example, our universe. The issue of an intractible likelihood is tackled with so-called simulation-based inference (SBI). The idea behind SBI is to train a NN on simulated data, were the inference parameters are known. Once the model is trained, inference is amortized: the model can perform inference on real data with the pre-trained model inexpensively. The prerequisite is, of course, that the model generalizes well from the training data to the real data.

SBI has been used to infer astrophysical and cosmological parameters [4], estimate the evolution of the neutral hydrogen fraction [5], and can also be used to test exotic cosmologies [6]. A prerequisite has also been the development of realistic simulators, such as 21cmFAST [7].More citations?

The issue of data processing raises the question of how much the 21 cm signal can be compressed, while the NN is still able to robustly constrain any relevant physical parameters. This can in parts be answered with the methods of information theory, which provides a framework to quantify the informativeness of the compressed data about parameters of interest. One is faced with the optimization problem of finding a maximally informative summary of the data, subject to the constraint that the compression is as stong as possible.

In this thesis, an architecture consisting of two neural networks is analyzed. The first NN is used to summarize simulated 21 cm data, while the second infers parameters from this summary. Another citation here? Two inference cases are tested, namely the reconstruction of the neutral hydrogen fraction during the EoR, and the recovery of the astrophysical and cosmological parameters used for the simulation.

2 Physics of the 21cm Signal

2.1 The cosmological setting

For the purposes of this thesis, a flat ΛCDM cosmology with an additional warm dark matter parameter is assumed. The ΛCDM-model is based on the Friedmann-Robertson-Walker metric:

$$ds^2 = -dt^2 + a^2(t)\left(\frac{dr^2}{1-kr^2} + r^2\big(d\theta^2 + \sin^2\theta d\phi^2\big)\right). \tag{1}$$

where $a(t)$ is the dimensionless scale factor and k is the curvature constant.

Combined with Einstein's field equations and the assumption that the universe may be described as a perfect fluid, the dynamics of this universe are described by the Friedmann equations:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{k}{a^2}, \tag{2}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3}. \tag{3}$$

where G is Newton's constant, Λ is the cosmological constant and ρ is the energy density of the universe. The scale factor as well as the energy density are time-dependent quantities.

Given the equation of state, $p = \omega\rho$, one finds for the density of different matter species:

$$\rho \sim \frac{\rho_0}{a^{3(1+\omega)}} \tag{4}$$

which yields the behavior of densities of

1. Radiation: $\rho_r \sim 1/a^4$
2. Matter: $\rho_m \sim 1/a^3$
3. Vacuum energy: $\rho_\Lambda \sim \text{const}$

Allowing matter, radiation and the cosmological constant to all be present, the Friedmann equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{\kappa}{a^2} = \frac{8\pi G}{3}(\rho_m + \rho_r) + \frac{\Lambda}{3}. \tag{5}$$

By defining the Hubble parameter

$$H = \frac{\dot{a}}{a} \tag{6}$$

and dividing 5 by H^2, one can identify the three terms on the r.h.s with energy densities:

$$1 + \frac{\kappa}{H^2 a^2} = \Omega_m + \Omega_r + \Omega_\Lambda \tag{7}$$

Knowledge of these density parameters therefore allows to determine the evolution of the scale factor a. Refer to conservation laws etc that allow to fix the densities

The ratio $H = \dot{a}/a$ is called the Hubble parameter and measures how fast the universe is expanding in the following sense: given two observers at $(t_1, r_1), (t_2, r_2)$ in a FRW-universe, where observer 1 emits a signal of duration Δt_1, observer 2 seeds a scale-factor dependent dilation of the signal duration, as measured by the cosmological redshift:

$$z = \frac{\Delta t_2}{\Delta t_2} - 1 = \frac{a(t_2) - a(t_1)}{a(t_1)} > 0 \quad \text{if universe expanding} \tag{8}$$

In the limit of $t_2 \to t_1$, if d is the instantaneous distance between the observers, one then has

$$z \approx \left.\frac{\dot{a}}{a}\right|_{t_1} d + \mathcal{O}((t_1 - t_2)^2) \tag{9}$$

such that H measures the expansion rate at a given time in units $1/\mathrm{s}$.

The Friedmann equations describe the global expansion history of the universe under strong symmetry assumptions. Their results match observations very well: citation They act as an effective theory though, since, for example, they contain no information about specific particles and their interactions with each other ausführen. Cosmology draws on other areas of physics to answer these questions. The following gives a short summary of the history of the universe when considering these as well.

Maybe just have a nice image of this long text below, it's taking up lots of space.

The first phase of the universe is marked by *radiation domination*. Immediately after the Big Bang, the universe underwent a period of rapid expansion called *inflation*, during which space expanded by appriximately an order of 10^{26}. This inflation was driven by the so-called inflaton field. During the following process of reheating, the energy of the inflaton field was converted to SM particles, which enabled subsequent *baryogenesis*, i.e. the creation of matter and antimatter particles. In the subsequent annihilation of matter.antimatter pairs, a small excess of matter paticles survived, composing the matter content of the universe today. There have been many proposed mechaninisms for both reheating and baryogensesis, and both are active fields of research. Insert sources here In the following time scale of $20 \cdot 10^{-12}\mathrm{s} - 1000\mathrm{s}$, particles acquired mass and coalesced into hadrons. Neurinos decoupled from baryons and thus the cosmic neutrino background was formed. What followed was an epoch named *big bang nuceosynthesis*, during which mostly protons and neutrons bound to primordial nuclei, consisting mostly of hydrogen, Helium-4 and traces of deuteron, Helium-3 and lithium.

The second phase of the universe is marked by *matter domination*. During *recombination*, electrons and nuclei where bound to neutral atoms. Photons were no longer in thermal equilibrium with matter and thus "decouple", forming the cosmic microwave background (CMB). Measurements of the CMB today provide one of the most crucial constraints on cosmological parameters.

After recombinations, most photons propagate freely and are quickly redshifted, such that the universe becomes dark. This period is therefore called the *dark ages*. Matter has not yet coalesced into stars and galaxies. The main source of photons visible phtons? is the 21 cm emission line of neutral hydrogen.

Once first stars and galaxies have formed, they emit photons energetic enough to ionize the surrounding hydrogen. This period is called the *epoch of reionization* (EoR). By the end of the EoR, most of the IGM is ionized and continues to be so today.

Some important cosmological params with definitions and their values nowadays, cite PDG

Table 1: Parameters of ΛCDM with estimated values

Name	Symbole	Value	Unit	Source
Hubble constant	H_0	67.4	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$	Planck 2018
Matter density	Ω_m	0.315 ± 0.007		Planck 2018
Dark energy density	Ω_Λ	0.685		Planck 2018
Spectral index	n_s	0.965		Planck 2018
Matter fluctuation amplitude	σ_8	0.811 ± 0.006		Planck 2018
Equation of state	w	-1		Planck 2018

Cite a few papers that give the current status of cosmology in ΛCDM, e.g. Planck 2018, etc.

Cite current values for all params that will be inferred

Need small intro on warm dark matter

How other cosmological models can be tested with the EorFlow setup, maybe also put this somewhere else.

What is matter distribution at the redshift when beginning study?

Small overview of existing studies of timeline of universe, why there is a gap in EoR

Maybe cite more recent paper that explains why studying this exact regime of redshifts

2.2 The origin of the 21cm signal

Need a source for this section.

Maybe some more details on IGM at these redshifts?

The two electron spin states of neutral hydrogen in the $1s$ state have slightly different energy levels. This hyperfine structure originates from the parallel (triplet state) and antiparallel (singlet state) orientation of electron spin with respect to the nuclues. The transition between states is forbidden, the excited state having a mean lifetime of $\sim 11 \cdot 10^6$ yrs. The triplet state carries slightly more energy, the energy difference being $\Delta E \approx 10^{-6}$ eV, which corresponds to a frequency of $\nu \approx 1.42$ GHz, or a wavelength of $\lambda \approx 21$ cm.

Neutral hydrogen gas made up most of baryonic matter during CD and EoR, and 21 cm photons emitted during these epochs can be measured today. A 21 cm photon emitted at some point during CD or EoR is redshifted by the time it reaches the earth. At redshift ranges $z \sim 6-30$, it is now measured on a range of $\nu \sim 46-200$ MHz, or $\lambda \sim 1.5-6.5$ m. This wavelength range belongs to the regime of radio astronomy and is covered by the SKA-Low setup, which is sensitive to frequencies $\nu \sim (50-350)$ MHz. Caroline's comment: lower than 6?. https://www.skao.int/en/science-users/118/ska-telescope-specifications

There are various mechanisms that lead to changes in this spin state of neutral hydrogen in the IGM. Those mechanisms that are consequences of the quantum mechanical properties of the atom are measured by Einstein coefficients, which are introduced in the following. The singlet state is denoted with index 0 and the triplet state with index 1. Then the index 01 denotes excitation, and 10 de-excitation. Indices in quotations?

A_{10} describes spontaneous emission and can be given as a rate. For neutral hydrogen, one finds $A_{10} \approx 2.85 \cdot 10^{-15}$ s^{-1}.

$B_{01/10}$ describes absorption/stimulated emission. For neutral hydrogen, one finds $B_{10} = A_{10}(c^2/2h\nu^3)$ and $B_{10}g_1 = B_{01}g_0$, where ν is the 21 cm line frequency given above and g is the spin degeneracy factor of each state. In terms of the 21 cm it is used to describe the interaction of neutral hydrogen clouds with CMB radiation.

Two other effects also lead to the (de-)excitation of neutral hydrogen in the IGM, and their rates are described by "effective" coefficients. C denotes those processes driven by collision interactions with other hydrogen atoms, free electrons and protons, and P those from scattering with Lyman-α photons. In equilibrium, both define a kinetic temperature T_K and effective color temperature T_c: How are they density dependent?

$$\frac{C_{01}}{C_{10}} = \frac{g_1}{g_0} e^{-\frac{T_*}{T_K}} \tag{10}$$

$$\frac{P_{01}}{P_{10}} = \frac{g_1}{g_0} e^{-\frac{T_*}{T_c}} \tag{11}$$

where again $T_K, T_c \gg T_*$ and the exponential can be expressed in first-order Taylor expansion.

On a large scale, the relative populations n_0 and n_1 of neutral hydrogen atoms in different spin states are described by (or determine) the spin temperature T_S:

$$\frac{n_1}{n_0} = \frac{g_1}{g_0} \exp\left(-\frac{E_{10}}{k_B T_S}\right) \equiv \frac{g_1}{g_0} \exp\left(-\frac{T_\star}{T_S}\right), \tag{12}$$

As a reminder, the g_i are spin degeneracy factors and E_{10} the energy difference between spin states. An estimate of T_S allows an estimate of the relative populations. For astrophysical environments, $T_S \gg T_*$ and as such $n_1 \approx 3n_0$ is a working approximation.

An estimate of T_S is then obtained by a rate balancing of the aforementioned Einstein coefficients:

$$n_1(A_{10} + B_{10}I_\gamma + C_{10} + P_{10}) = n_0(B_{01}I_\gamma + C_{01} + P_{01}) \tag{13}$$

About thermal equilibrium: quote from 21 cm text: Note that the relevant timescales are all much shorter than the expansion time, so equilibrium is an excellent approximation.

where I_γ is the specific intensity of CMB photons at the frequency of the 21 cm line always saying line.... The CMB background is the source closest resembling a blackbody spectrum, which is in this frequency range described by the Rayleigh-Jeans law, with which I_γ can be written in terms of a temperature at the frequency of the 21 cm line. Plugging everything into 13 yields the following approximation to T_S:

$$T_S^{-1} = \frac{T_\gamma^{-1} + x_c T_K^{-1} + x_\alpha T_C^{-1}}{1 + x_c + x_\alpha} \tag{14}$$

$$x_c = \frac{C_{10}}{A_{10}} \frac{h\nu}{k_B T_\gamma} \tag{15}$$

$$x_\alpha = \frac{P_{10}}{A_{10}} \frac{h\nu}{k_B T_\gamma}, \tag{16}$$

This raises the question of what the measured 21 cm looks like, given this estimate of T_S.

The 21 cm signal is given in terms of brightness offset temperature δT_B which is calculated from the specific intensity measured by the radio telescope, assuming the Rayleigh-Jeans regime. Let ν be the frequency at which the 21 cm signal is to be observed. The signal is then composed of the background CMB radiation and the radiation emitted from clouds of neutral hydrogen, minus the CMB radiation absorbed in the neutral hydrogen cloud. This is expressed as

$$\frac{dI}{ds} = -\alpha(\nu)I_\gamma + \alpha(\nu)I_S \tag{17}$$

where I_γ denotes the CMB specific intensity, I_S that of the emission of the 21 cm signal from the neutral hydrogen cloud and $\alpha(\nu)$ the absorption coefficient of neutral hydrogen.

The observered specific intensity I_B is obtained by solving the radiative transfer equation from 17, yielding:

$$I_B = I_S(1 - e^{-\tau}) + I_\gamma e^{-\tau} \tag{18}$$

$$\tau(\nu) = \int \alpha(\nu) ds \tag{19}$$

where τ is the so-called optical depth. This carries the interpretation that for an optically thick material, $\tau >> 1$, the signal is dominated by the 21 cm signal and for an optically thin material, $\tau << 1$, by CMB radiation.

In the Rayleigh-Jeans regime, this observed specific intensity is proportional to the temperature. Adding the redshift correction for the CMB temperature, this yields

$$T_{B0}(1+z) = T_S(1-e^{-\tau}) + T_0(1+z)e^{-\tau}. \tag{20}$$

Finally, the brightness offset temperature is obtained by subtracting the CMB background:

$$\delta T_B = T_{B0} - T_0 = \frac{T_S - T_\gamma}{1+z}(1-e^{-\tau}). \tag{21}$$

The quantities T_S and τ have to be estimated. In practice, this is done numerically. In the following are some comments on the procedure.

Now, τ has to be estimated. The absorption coefficient measures the probability of an interaction per frequency and unit distance. Given an Einstein coefficient X, it is given by

$$\alpha(\nu) = \frac{h\nu}{c} X n \phi(\nu), \tag{22}$$

where n is number density of atoms partaking in the interaction and ϕ is the line profile normalized to one. Then one finds for neutral hydrogen:

$$\alpha(\nu) = \frac{h\nu}{c}(B_{01}n_0 - B_{10}n_1)\phi(\nu) \tag{23}$$

$$\approx \frac{h\nu}{c} n_0 B_{01} \frac{h\nu}{k_B T_S}\phi(\nu), \tag{24}$$

the approximation coming from equation 12. The line profile is approximated by $\phi(\nu) \approx c/(sH(z)\nu)$ define variable names, an estimate made when considering the effect of cosmological recession. Then

$$\tau(\nu) = A(\nu)\frac{x_{HI} n_H}{T_S H(z)} \tag{25}$$

$$A(\nu) = \frac{3c^2 h A_{10}}{32\pi\nu^2 k_B}. \tag{26}$$

Assuming matter domination yields $H(z) = H_0 \Omega_{m,0}^{1/2}(1+z)^{3/2}$ with H_0 and $\Omega_{m,0}$ the present-day cosmological constant and matter density parameter What about n_H?, one obtains a more detailed approximation of the brightness offset temperature from 21 in

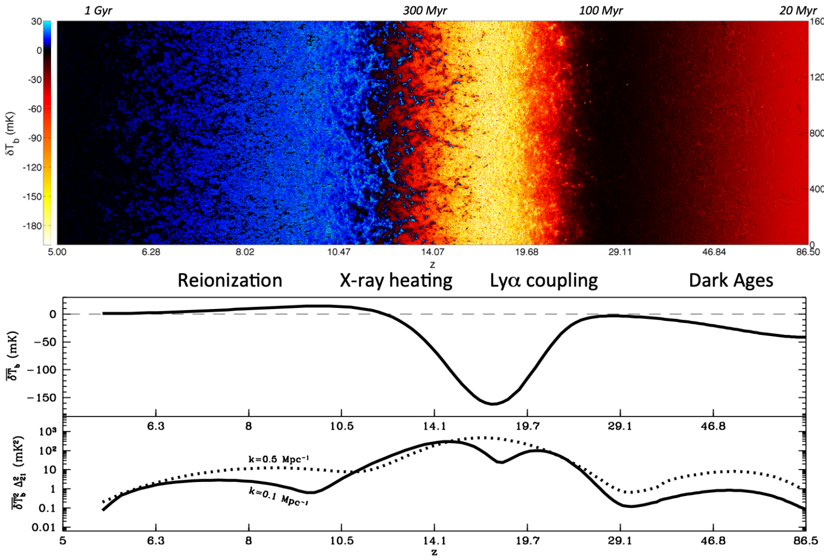


Figure 1: A map of the 21 cm signal of neutral hydrogen

the limit $\tau \ll 1$:

$$\delta T_B \approx 27\, x_{\mathrm{HI}}(1+\delta_b)\left(1-\frac{T_\gamma}{T_S}\right)\left(\frac{1+z}{10}\,\frac{0.15}{\Omega_{m,0}h^2}\right)^{1/2}\left(\frac{\Omega_{b,0}h^2}{0.023}\right)\,\mathrm{mK} \tag{27}$$

with δ_b the fluctuation of baryon content, $\Omega_{b,0}$ the present-day baryon density and x_{HI} the neutral hydrogen fraction.

In practice, any realistic simulation must also take into account foreground signals such as from astrophysical sources or instrument noise. Noise modelling then becomes an additional challenge.

2.3 Qualitative behavior of the 21cm signal

From the expression in equation 27, it becomes clear that if $T_S > T_\gamma$, the measured brightness offset is positive and vice versa.

Will still have some comments on the figure

3 Machine Learning Background

As an introduction to the methods used in this field, this section covers the concept of neural posterior approximation, some theory behind the class of models analyzed in this thesis as well as a summary of metric from information theory, and how they can evaluate a model's quality.

In the following, x denotes a data sample, and θ a corresponding parameter. Probability densities are denoted p, and their approximations with q. Functions s denote objects that summarize data, so $s(x)$ is such a summary. Neural networks are denoted with an index $\phi or \psi$, so q_ϕ is an estimate of a probability distribution done with a neural network, and s_ψ is a summary done by a neural network.

3.1 Neural posterior estimation

In a Bayesian inference task, processing these data would typically be done by choosing:

1. a likelihood $p(x|\theta)$ describing how observed data arised from a specific set of parameters,
2. a prior $p(\theta)$ describing beliefs about the parameter distribution before seeing data.

Given data d, one then calculates the posterior distribution using Bayes' theorem:

$$p(\theta|x) \sim p(x|\theta)p(\theta). \tag{28}$$

In many applied cases, an analytic shape of the likelihood is not available. An example is when the data come from a complex simulator. Then one can sample from this simulator and as such from the likelihood, but the expression $p(x_{\text{observed}}|\theta)$ is not obtainable and as such, the posterior cannot be calculated. In such setting, an alternative is to model the posterior separately, such that given x_{observed}, one can at least sample from the posterior: $\theta \sim q(\theta|x_{\text{observed}})$. This is known as likelihood-free inference. When the posterior is approximated using a neural network $q_\phi(\theta|x)$, one also speaks of neural posterior estimation (NPE).

An advantage is that unlike with the traditional Bayesian methods, the posterior does not have to be re-calculated given new data: the neural network simply has to be evaluated on the new dataset. In this way, the cost of obtaining the posterior was payed beforehand during training. This concept is known as amortized inference. In order to train the model, data is needed, and this is where simulators are implemented. The posterior model is trained on simulated data and once trained, one can sample parameters by $\theta \sim q_\phi(\theta|x_{\text{observed}})$. In this way, NPE becomes a way of implementing simulation-based inference.

3.2 Information Theory and Neural Estimators

Traditionally, data as complex as those expected from SKA are analyzed using physics-inspired summary statistics $s_{\text{phys}}(x)$, which map the data to a lower-dimensional space with known interpretation. An example is the power spectrum. Such a summary comes with the benefit of interpretability, but the caveat is a potentially large information loss.

This motivates the use of a neural networks to learn a different summary $s_\phi(x)$, and the procedure is then called representation learning. The summarry is hopefully more informative of the parameter of interest, though potentially less interpretable. If $\mathbb{I}$ is a measure of informativeness between the data and the parameters, this increased informativeness can be expressed as:

$$\mathbb{I}(\theta; x) \geq \mathbb{I}(\theta; s_\phi(x)) > \mathbb{I}(\theta; s_{\text{phys}}(x)). \tag{29}$$

The necessitates a measure of informativeness for use in representation learning, which is introduced in this section.

Some Basic Measures in Information Theory

Given a random variable X distributed as p, $X \sim p$, the entropy of X is defined as:

$$\mathbb{H}(X) = -\mathbb{E}_X \left[\log p(X)\right] \geq 0. \tag{30}$$

If the logarithm is chosen with base 2, one defines the unit of this quantity as bits. If one uses the natural logarithm, the entropy is measured in units of nats.

Entropy is a measure of the uncertainty of a random variable. A high entropy corresponds to a lack of predictability of the random variable's outcome. As an example, the entropy of the uniform distribution is maximal, while that of the delta distribution is zero. One therefore considers entropy to be a measure of the amount of information contained in the random variable's distribution.

The generalization to conditional entropy is:

$$\mathbb{H}(X|Y) = -\mathbb{E}_{X,Y} \left[\log p(X|Y)\right]. \tag{31}$$

Another important metric in information theory is the Kullback-Leibler divergence. Let $X \sim p$ and $Y \sim q$ be two random variables over the same sample space. The KL-divergence between p and q is defined as:

$$\mathbb{D}_{\text{KL}}(p||q) = \mathbb{E}_X \left[\log p(X) - \log q(X)\right] \geq 0. \tag{32}$$

The KL-divergence satisfies the identity axiom, $\mathbb{D}_{\text{KL}}(p||q) = 0 \Leftrightarrow p = q$. It is not symmetric and does not satisfy the triangle inequality and as such is not a metric. However, it does still carry an interpretation as a distance between two probability distributions: from the above expression it is clear that the more q resembles p, the smaller the measured KL-divergence becomes. A smaller the KL-divergence between the two distributions means there is less information loss by using q to approximate p.

Since the KL-divergence is assymmetric, one distinguishes $\mathbb{D}_{\mathrm{KL}}(p||q)$ as the forwards KL between p and q, and $\mathbb{D}_{\mathrm{KL}}(q||p)$ as the backwards KL between p and q. Minimizing the forwards KL-divergence between the data distribution and the model distribution is equivalent to minimizing the negative log-likelihood (NLL), because

$$\mathbb{D}_{\mathrm{KL}}(p_{\mathrm{data}}||q_\phi) = -\mathbb{H}(p_{\mathrm{data}}) - \mathbb{E}_{x\sim p_{\mathrm{data}}}\left[\log q_\phi\right] = \mathrm{NLL} + \mathrm{const.} \tag{33}$$

Lastly, the mutual information between two random variables X and Y is defined as:

$$\mathbb{I}(X;Y) = \mathbb{D}_{\mathrm{KL}}(p(X,Y)||p_X(X)p_Y(Y)) \geq 0. \tag{34}$$

An alternative formulation in terms of entropy is:

$$\mathbb{I}(X;Y) = \mathbb{H}(X) - \mathbb{H}(X|Y) = \mathbb{H}(Y) - \mathbb{H}(Y|X). \tag{35}$$

Mutual information is symmetric and nonnegative. It is zero if and only if X and Y are independent and becomes maximal if and only if $X = Y$. One interprets its value as the amount of information obtained about one random variable while observing another. Since it is defined as an expectation over the joint distribution of the two variables X and Y, it is sensitive to higher moments of these distributions, and cannot be captured simply be e.g. their variance.

Mutual Information Estimation

In the context of representation learning, mutual information can be used as a measure of model quality. This is motivated by the data processing inequality, which in this context can be phrased as follows:

Let X, Y and Z denote random variables. These are processed as $X \to Y \to Z$, in the sense that a sample from X is processed to produce a sample from Y, which in turn is processed to produce a sample from Z, without any additional knowledge of X. Then Z cannot contain more information about X than Y, so

$$\mathbb{I}(X;Y) \geq \mathbb{I}(X;Z). \tag{36}$$

This applied in the context of SBI and NPE, where the parameters of interest θ are part of the first random variable, the simulated data x part of the second and the summarized data $z = s_\phi(x)$ part of the third. The data processing inequality then reads:

$$\mathbb{I}(\theta;x) \geq \mathbb{I}(\theta;z), \tag{37}$$

and this explains why mutual information is the measure of informativeness mentioned in the beginning of this section. A good summary keeps as much information about the original data as possible, subject to the constraint that it should contain less total information than the data itself, i.e. summarize it in some way. The part of the network in which the compression of data is the highest is often referred to as the bottleneck.

Something on information bottleneck method?

In the setting of NPE, one can use the learned posterior to estimate mutual information between the bottleneck and the parameter distributions in a natural way. Even though the mutual information between the original data and the parameters remains inaccessible without a separate network, this allows the comparison of different summary networks and their informativeness about the parameters. The mutual information between the bottleneck and the parameter distributions is defined as

$$\mathbb{I}(\Theta; Z) = \int p(\theta, z) \log \frac{p(\theta|z)}{p(\theta)} \,, \tag{38}$$

where the distribution $p(\theta|z)$ is now estimated by a learned $q_\phi(\theta|z)$, the optimization objective being to minimize the KL-divergence $\mathbb{D}_{\mathrm{KL}}(p||q_\phi)$. The Monte-Carlo approximation of eq. 38 using this distribution is referred to as the Barber-Agakov (BA) estimate [8]:

$$\hat{\mathbb{I}}(\Theta; Z) = \mathbb{I}_{\mathrm{BA}}(\Theta; Z) = \mathbb{E}_{P(\Theta,Z)} \left[\log q_\phi(\theta|z) - \log p(\theta)\right]. \tag{39}$$

This is computable, because the distribution of the prior $p(z)$ can in practice either be ignored as a constant factor, or in case of SBI is chosen and therefore known. If this is not the case, $p(z)$ can also be estimated separately, which however introduces more uncertainty in the above estimate.

Because the KL-divergence is non-zero, the BA-estimate yields a lower bound on the true mutual information:

$$\mathbb{I}(\Theta; Z) \geq \mathbb{I}_{\mathrm{BA}}(\Theta; Z). \tag{40}$$

Qualitatively, one expects mutual information to behave as follows: If the parameters being inferred are completely independent of one another, a perfect summary network would still require a bottleneck of the same dimensions as the parameter space in order to correctly infer the parameters. For lower-dimensional bottlenecks, one therefore expects an information loss. For higher-dimensional bottlenecks, one expects a saturation in mutual information, since the network does not require more dimensions in order to learn more about the required parameters. In practice, given noisy and imperfect data, realistic network performance and the fact that the BA-estimate only yields a lower bound of mutual information, the behaviour might be more ambiguous. In general, the network likely requires more dimensions to learn an informative summary, especially due to noise. If the parameters of interested are correlated, however, less dimensions might be required.

3.3 An architecture for complex data

An architecture that has proven to be successful in recent SBI approaches (see [4], [5] or [6]) combines two neural networks in the following principle:

1. Image data are used as input for a convolutional neural network (CNN), which compresses these into a much lower-dimensional summary.
2. This summary is mapped to whichever feature is being inferred by a conditional normalizing flow (CNF).

The idea is that the CNN learns to efficiently encode any relevant physics information, while the flow learns to interpret the CNN's output and map it to the inferred quantity. These two networks are introduced in the following.

Concerning notation: The summation convention is used and the bias term is implicit. For example, a multilayer perceptron (MLP) would be described as follows: A basic MLP consists of only one type of layer. If h is the output of the previous layer, the current layer is defined by

$$z_i = \phi(W_{ij} h_j), \tag{41}$$

where ϕ will be used to denote an activation function and W is the weight matrix. This type of pre-activation $W_{ij} h_j$ (notice implicit bias!) is also known as an affine transformation.

Convolutional Neural Networks

CNNs are inspired by the task of implementing neural networks in image analysis, i.e. on data with 2D spatial structure. To explain the novel concepts behind the CNN in comparison with a MLP, the CNN architecture will be discussed in 1D. The generalization to higher dimensions is straightforward.

Let the input data have dimensions $x \in \mathbb{R}^m$. The CNN learns a function $s_\phi(x)$ which summarizes the data. It multiple types of layers to achieve this:

A convolutional layer applies a convolution operaton:

$$z_i = \phi([w \circledast h]_i) \equiv \phi\left(\sum_k w_k h_{i+k}\right) \tag{42}$$

where the dimensionality of the kernel is fixed $w \in \mathbb{R}^K$ is typically much lower than that of the data. Since the kernel is used on all parts of the previous layer, it can learn features in a translation-independent way.

A pooling layer applies a pooling operation:

$$z_i = \text{pool}\, K_{ij} h_j \tag{43}$$

where K defines a pooling window for each index i. This results in neighboring points of h_j to be included in the operation. Typical choices for pool are maxpool, which chooses the maximum of all given values, or meanpool, which calculates their mean. The pooling layer downsamples the previous layer and enables an extraction of features independent of length scales.

During construction of the CNN one needs control of the geometry of the data in each layer, and a number of additional parameters are introduced to achieve this. A padding parameter p controls the border of the output layer by describing how the kernel or pooling window interacts with the previous layer's border. A stride parameter s allows the kernel or pooling window to operatore along the input layer by skipping every s'th pixel. In order to enable the CNN to learn more features one also typically adds more channels to the data as it flows through the CNN, and each channel is associated with it's own weights.

A typical CNN architecture consists of multiple interchanging convolutional and pooling layers, after which the output is flattened to a vector and connected by a few dense layers to the final output vector. An example is shown in fig 2. The final CNN takes an input $x \in \mathbb{R}^{C\times H\times W}$, where C is the channel dimension, H the height dimension and W the width dimension, and learns a summary $f(x) \in \mathbb{R}^n$. For a simple supervised learning task, the CNN could be trained on an MSE loss to recover parameters originally associated with the image.

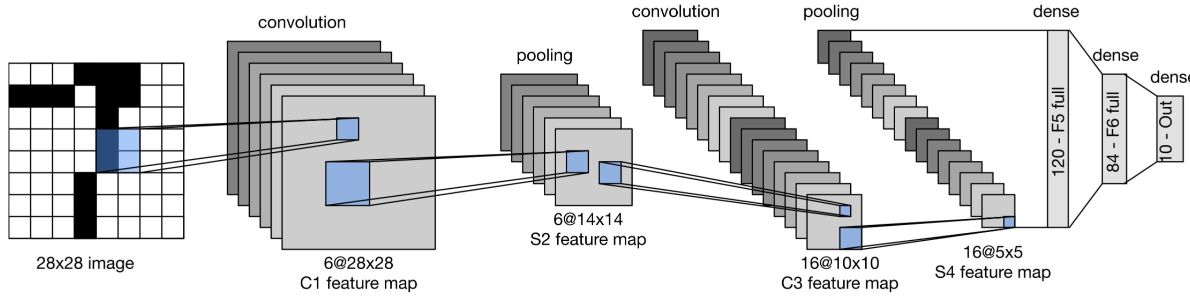


Figure 2: Tpical CNN architecture, taken from [9]

Conditional Normalizing Flows

Need to agree on how sampling is written for expectation

A normalizing flow is a model that learns to transform a simple (often Gaussian) base distribution into a more complex one, the target distribution. The transformation is chosen to be invertible, such that the model becomes generative: once trained, it can be inverted and by transforming samples from the normal, one can sample from the approximated target distribution. The following is based on [10].

Let $u \sim p_u = \mathcal{N}(0, I_n)$ with $n \in \mathbb{N}$ the base distribution and $\theta \sim p_\theta$ with p_θ the n-dimensional target distribution. Let $f : \mathbb{R}^n \to \mathbb{R}^n$ a diffeomorphism, with $g = f^{-1}$ its inverse. The training objective is to find g, such that

$$u = g(\theta) \sim p_u, \qquad \theta \text{ sampled from } p_\theta. \tag{44}$$

The process of generating samples from the approximated distribution is the inverse of the above:

$$\theta = f(u) \sim q_\theta, \qquad u \text{ sampled from } p_u. \tag{45}$$

For a given transformation, one computes $p_\theta(\theta)$ by applying the change of variables formula:

$$q_\theta(\theta) = p_u(g(\theta)) \left| \det \left(\frac{\partial g(\theta)}{\partial \theta} \right) \right| = p_u(u) \left| \det \left(\frac{\partial f(u)}{\partial u} \right) \right|^{-1}. \tag{46}$$

One optimizes the flow by minimizing the forwards KL-divergence between the true and the approximated distribution, which as discussed ins section 3.2 is equivalent to minimizing the NLL:

$$\mathcal{L}_{\mathrm{NF}} = -\mathbb{E}_{\theta \sim p_\theta} \left[\frac{|g(\theta)|^2}{2} + \log \left| \det \left(\frac{\partial g(\theta)}{\partial \theta} \right) \right| \right] \tag{47}$$

where the first term stems from the choice of a Gaussian latent space and the second from the jacobian in eq. 47.

The normalizing flow becomes expressive by allowing the transformation to be as complex as possible, while restricting it to a class of functions with efficiently computable Jacobian determinants makes it computationally tractable.

4 Implementation

Need specific values of learning rate etc for both models. Should this go in appendix?

4.1 The dataset

For this project, an existing dataset simulated with 21cmFast [7] is used. The simulation parameters and their priors are shown in table 2. Parameters are sampled uniformly from their priors for simulations. All other parameters of the flat ΛCDM cosmology are fixed to the Planck measurements [1]. This means the baryon density is $\Omega_b = 0.04897$, the matter fluctuation amplitude is $\sigma_8 = 0.8102$, the dimensionless Hubble parameter is $h = 0.6766$, and the scalar spectral index is $n_s = 0.9665$.

Table 2: Parameters and priors used for the simulations

Name	Symbol	Prior
Matter density	Ω_m	$\mathcal{U}(0.2, 0.4)$
Warm dark matter mass	ω_{WDM}	$\mathcal{U}(0.3, 10)$ keV
Minimum virial temperature	T_{vir}	$\mathcal{U}(10^4, 10^{5.3})$ K
Ionization efficiency	ζ	$\mathcal{U}(10, 250)$
Specific X-ray luminosity	L_X	$\mathcal{U}(10^{38}, 10^{42})$ erg/s/$\text{M}_\odot \times$ yr
X-ray energy threshold for self-absorption by host galaxies	E_0	$\mathcal{U}(100, 1500)$ eV

Each of these parameters can be described as follows:

- The matter density Ω_m, describing the fraction of the universe's energy density from matter (incl. dark matter)
- The warm dark matter mass ω_{WDM}, describing the mass of a hypothetical warm dark matter particle
- The minimum virial temperature T_{vir}, describing the minimum virial temperature to enable star formation in the first places
- The ionization efficiency ζ, characterizing the strength of ionizing radiation
- The specific X-ray luminosity L_X, a measure of total X-ray luminosity of galaxies, measured per star formation rate for comparability
- The X-ray energy threshold for self-absorption by host galaxies E_0, a measure for the energy of X-rays below which these do not escape their host galaxies.

Could still comment on the significance of each parameter. Why are X-rays so relevant?

The final dataset consists of 5000 sets of lightcones with the corresponding simulation parameters θ and an array of neutral hydrogen fractions x_{HI}. Each lightcone is originally

$140 \times 140 \times 2350$ voxels in size, corresponding to a redshift range of $z \sim 5 - 35$ and a box size of 200 Mpc, such that the resolution is 1.43 Mpc. Since Ω_m changes the redshift range captured with the first 2350 pixels of the simulated lightcone, this redshift range corresponds to the maximal redshift range a lightcone encompasses. The neutral hydrogen fractions are stored as 92-dimensional vectors. The distribution of these x_{HI}-labels is not uniform. To impose a realistic constraint, lightcones are only admitted to the dataset if the optical depth τ is within 5σ of the Planck [1] measurement and the IGM mean neutral fraction $\hat{x}_{\mathrm{HI}}$ is below 0.01 at redshift $z = 5$.

The dataset is preprocessed as follows: The temperature offset values are min-max-normalized individually for each lightcone, which later improves the CNN's performance. The lightcones are truncated to $140 \times 140 \times 1916$ voxels. This allows the neutral hydrogen fraction labels to first be truncated to 75 dimensions and then binned to 15-dimensional vectors, reducing the dimensionality of the inference problem. An additional logit transformation is applied to this vector, which improves training performance of the flow. A sigmoid transform can be applied during inference to map the labels back to their physical state. The simulation parameters, which span many different orders of magnitude, are min-max normalized globally. The total dataset has a size of approximately 690 GB and is split into a training, validation and test set with relative sizes 0.7, 0.2 and 0.1, respectively.

4.2 The network architecture

Section 3.3 the basic architecture used in the inference task. In this specific implementation, a lightcone is passed through the CNN, who's output is used as a conditional input to the CNF model to infer either the original simulation parameters or the neutral hydrogen fraction. The general architecture is visualized in 3.

The conditional input changes the loss function of the flow from eq 47. Let x denote an image from lightcone distribution, θ the corresponding element of the label distribution, s_ψ the CNN and g_ϕ the transformation defining the normalizing flow. The network's parameter indices are surpressed in the following. For both inference cases, the loss function for jointly training the two models is:

$$\mathcal{L}_{\mathrm{NPE}} = -\mathbb{E}_{(x,\theta)\sim p(x,\theta)} \left[\frac{|g(\theta|s(x))|^2}{2} + \log \left| \det \left(\frac{\partial g(\theta|s(x))}{\partial \theta} \right) \right| \right]. \tag{48}$$

The architecture for inference of the original simulation parameters is based on [6], and that for inference of the reionization history on [5]. Table 3 details the architectures for both inference cases. In the CNN, the first convolutional layer is designed to learn

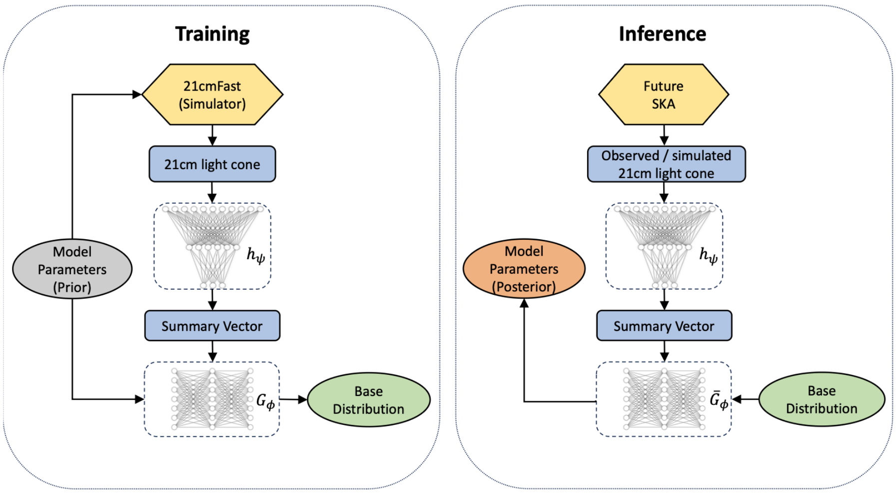


Figure 3: General architecture, taken from [6]

patterns in the redshift direction. An aggressive stride is used to alter the dimensionality of the next layers. The kernel for hidden layers is set to (3x3x2) and max pooling is only applied in spacial dimensions. Finally, all information is compressed into a 128-dimensonal feature vector by global average pooling. This is in turn transformed into an N_B-dimensional bottleneck vector by dense layers. In case the CNN is pretrained on the simulation parameters, an additional two dense layers can be attached, which map the N_B-dimensional output to a 6-dimensional output. These can be removed for the rest of training, leaving the rest of the architecture untouched. The CNF is used in two variants depending on the inference task, the heavier of the two architectures containing more coupling layers and more nodes per layer.

4.3 Setup for inferece of the neutral hydrogen fraction

For inference of the neutral hydrogen fraction, the CNN and CNF are trained jointly. Both models are trained using an optimizer, a scheduler and a joint scaler. The batch size is chosen as 8 and gradient accumulation to an effective batch size of 32 is implemented. The learning rate of the flow is initialized as 0.001 and that of the CNN as 0.0005. The optimization method is AdamW [11], a scheduler reduces the learning rate when the training loss stagnates and gradient scaling ensures more stable backpropagation. Training stops after 250 epochs.

Unlike the original distribution of simulation parameters, the prior distribution of reionization history vectors x_{HI} is not uniform. It therefore has to be modeled separately. It is modeled with a flow of the same architecture as the one from the conditional distribution, without any conditional input. An additional prior model adds uncertainty to the mutual information estimation.

Table 3: CNN network architecture

Layer	Shape
Input Layer	(140,140,1916,1)
3x3x75 Conv3D	(138,138,23,32)
3x3x2 Conv3D	(136,136,22,32)
2x2x1 Max Pooling	(68,68,22,32)
3x3x2 Conv3D	(66,66,21,64)
1x1x0 Zero Padding	(68,68,21,64)
3x3x2 Conv3D	(66,66,20,64)
2x2x1 Max Pooling	(33,33,20,64)
3x3x2 Conv3D	(31,31,19,128)
1x1x0 Zero Padding	(33,33,19,128)
3x3x2 Conv3D	(31,31,18,128)
Global Average Pooling	(128)
Dense	(128)
Dense	(128)
Dense	(128)
Dense	(N_B)
(Dense)	(128)
(Dense)	(128)
(Dense)	(6)

Table 4: Flow network architecture

Light network variant

Feature	Shape
Input layer	N_B
Coupling layers	8
Nodes	256
Subnet depth	3

Heavy network variant

Feature	Shape
Input layer	N_B
Coupling layers	10
Nodes	512
Subnet depth	3

4.4 Setup for inference of the simulation parameters

For inference of the original simulation parameters, training is done with AdamW optimization and a scheduled learning rate. No gradient scaling is needed. Using gradient accumulation with an effective batch size of 32 causes training to fail, therefore gradient accumulation is not implemented. The model does not converge if the CNN and flow are trained jointly from the beginning, so a three-step training process is implemented. First, the CNN is pretrained on the labels with an MSE loss. Since this has to be done for varying bottleneck dimensions, the additional two dense layers are added. Pretraining ends after 50 epochs. The CNN weights are then frozen and the dense head is removed for the rest of training. The CNN outputs after pretraining are then used to pretrain the flow for 120 epochs. Afterwards, the CNN weights are unfrozen and the optimizers and schedulers of both models are reset. Both models are then trained jointly (finetuned) for another 80 epochs. The original learning rate for CNN pretraining is 0.0004, that for flow pretraining is 0.001 and the same learning rates halved are used for the reinitialized optimizers during finetuning.

5 Results and Discussion

Will have a few sentences on general results here

Training is performed on one NVIDIA A30 GPU.

Might be nice to evaluate some metric for inference performance that Yannic used as well, to show that I recovered the results.

A word on bootstrap error calculations for comparison

Training is performed on one NVIDIA-A30 GPU. Due to long training times and therefore expensive training, only specific bottleneck dimensions are analyzed. Along with varying bottleneck dimensions, a number of different weight initializations (seeds) are tested. One trained model and its resulting mutual information estimates are referred to as one sample. One batch of models with same bottleneck dimension but different seeds is referred to as one ensemble.

For both inference cases, training is done for 250 epochs, after which every observed model has reached it's best validation performance, begins to overfit and plateaus in both training and validation losses. This raises the question of which epoch is to be chosen for the evaluation of mutual information. Three possible decision rules are tested:

1. 5% rule: The model for which training and validation loss are last within 5% of each other.

2. Best validation rule: The model with the best validation performance.
3. Last epoch rule: The last epoch the model was trained on before training ends.

These rules are justified as follows: The 5% rule allows to choose a model based on training performance while preventing overfitting. The exact value of allowed relative difference is of course an additional hyperparameter. The best validation rule is standard practice in choosing a model after training and the hope is that such a model performs best on the test set as well. The last epoch rule can be justified retrospectively: Because the training process for this model is quite instable before it reaches it's plateau, with individual models performing significantly better or worse on the validation set, it allows to choose the model from the most stable part of the training process.

The final estimates on the conditional entropy as well as the mutual information are done on the held-out test set consisting of 10% of the total dataset (500 samples), according to:

$$\hat{\mathbb{H}}_C = -\mathbb{E}_{p(l,s)}\left[\log q_\theta(l|s)\right] \tag{49}$$

$$\hat{\mathbb{I}} = -\mathbb{E}_{p(l,s)}\left[\log q_\theta(l|s) - q_\phi(l)\right] \tag{50}$$

Compared to the rest of training, these metrics is inexpensive to evaluate, so it is done every epoch, which enables the visualization of estimated CE and MI during training. In the case of the inference of the neutral hydrogen fraction, the separate estimate of the prior leads to an additional bias in the final MI estimate. For this reason, the interpretation of results is preferably done on the estimates of CE, though an estimate of MI is also always given.

Two sources of error in the final estimations are considered. The error of the individual model, stemming from the variability of log probability values on the test set, is referred to as the *model error*. This is estimated from the standard error and confirmed by bootstrap analysis, see below. When comparing results across seeds, an additional *seed error* must be considered, which is also estimated from the standard error.

5.1 Estimates for models of the neutral hydrogen fraction

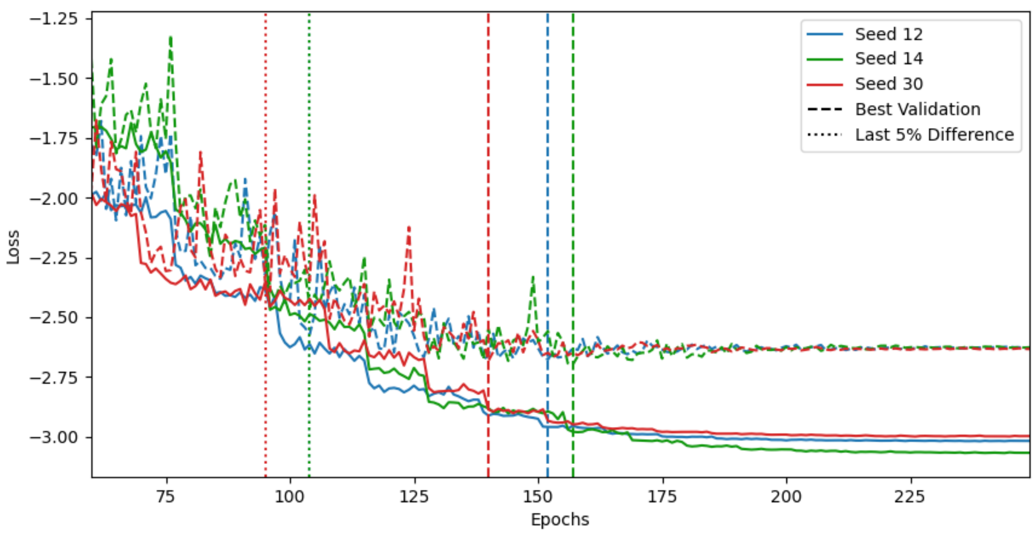
When evaluating CE and MI on models based on the 5% rule, the expected behavior of these metrics is in most parts recovered: The estimates of CE and MI increase with larger bottleneck dimension N_B, with the exception of an outlier model for $N_B = 8$. The other selection rules do not show this behavior.

Training the model

The average training time for this model is insert. The bottleneck dimensions analyzed are $N_B \in \{4, 6, 8, 15, 16, 26, 30\}$, each for two or three different seeds. As an example, the training and validation losses for an ensemble with bottleneck dimension $N_{\text{dim}} = 15$ are shown in figure 4. Vertical lines indicate the epoch chosen by best validation rule and dotted lines indicate the epoch chosen by the 5% rule. The loss plots for all other ensembles can be found in the appendix Reference coming soon.

In all cases, training and validation losses improve while still improving begin to diverge. Soon thereafter the best validation loss is reached and all models then reach a plateau in both training and validation loss. This behavior is observed for all architectures trained on the neutral hydrogen fraction. It is important to note that training of the ensemble for $N_{\text{dim}} = 15$ is comparatively stable, with training and validation losses converging to very similar values across seeds. This indicates that the model reaches similar regions in the loss landscape, independent of the inizialization. This is not the case for all models, and training for smaller bottlenecks in particular is less stable.

Figure 4: Train and Validation Losses for $N_B = 15$



Comparing model selection rules

In order to compare the model selection rules, table 5 summarizes the average amount of epochs it takes for a model to reach the epochs chosen by the 5% and best validation decision rules. Choosing models by the 5% rule results in a higher variance in the selected epoch for models with larger bottleneck dimensions, while choosing models by the best validation rule results in a higher variance in the selected epochs for models with smaller bottleneck dimensions. The 5% decision rule imposes the strictes decision choice, while model choice by best validation performance allows the model to learn longer before making a CE/MI estimate. For both selection rules, architectures with $N_B = 4$ train the longest before being chosen, but there is no strong pattern for the choice of model for the other N_B.

Table 5: Epochs selected for different selection rules

	5% rule	Best validation rule
$N_B = 4$	129 ± 21	174 ± 25
$N_B = 6$	98.5 ± 0.5	162 ± 5
$N_B = 8$	126 ± 7	169 ± 17
$N_B = 15$	102 ± 4	151 ± 7
$N_B = 16$	103 ± 25	156 ± 6
$N_B = 25$	116 ± 19	173.5 ± 0.5
$N_B = 30$	$109 \pm -$	$144 \pm -$

Estimates of conditional entropy and mutual information

In order to visualize how conditional entropy and mutual information are determined for a specific model, figure 5a shows a histogram plot of conditional probabilities of the test set, and 5b shows the probabilities from the separetely trained prior model. The estimate of conditional entropy and its error is then the mean and standard error of all $\log q_\phi(\theta_i|z_i)$, and the estimate of mutual information is obtained in the same way once all $\log q_\psi(\theta_i)$ have been subtracted. These results are marked in the respective figures.

(a) Estimation of conditional entropy

(b) Estimation of mutual information

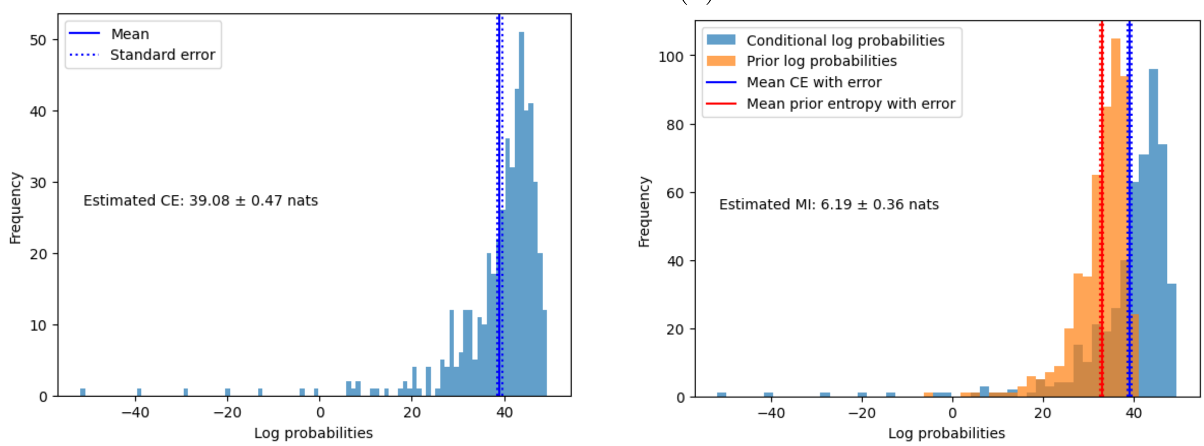
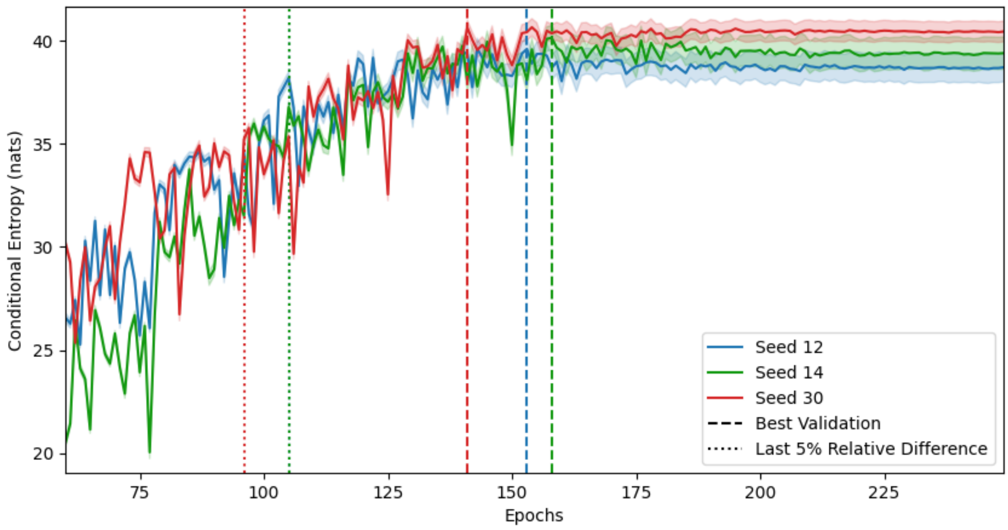


Figure 5: Histogram plots used to evaluate CE and MI

In this way, an estimate of conditional entropy and mutual information is obtained for every epoch. As an example, the estimated conditional entropy per epoch is shown in figure 6. Again, vertical lines indicate the epoch chosen by best validation rule and dotted lines indicate the epoch chosen by the 5% rule. The epochs chosen by 5% rule for seeds 12 and 30 coincide. All models reach a saturated estimate of the conditional entropy once the model plateaus in its training and validation performance. The standard error of the CE estimate is larger in the training phase during which the model begins to overfit. This qualitative behaviour is observed for all architectures and it is the same for estimates of MI as well. All resulting plots can be found in the appendix reference appendix.

Figure 6: Conditional Entropy per Epoch for $N_B = 15$



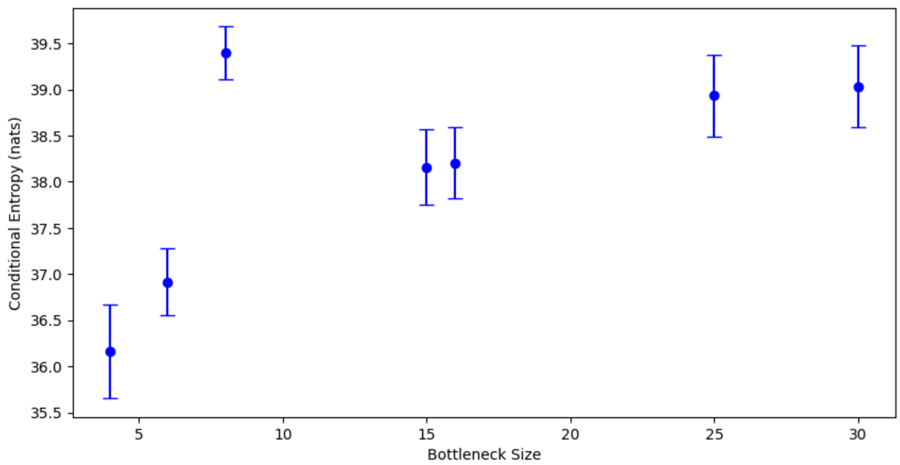
The estimates of CE and MI are done for all ensembles and for all selection rules. As an example, the results of final CE and MI estimates with their associated model and seed errors based on the 5% selection rule are shown in table 6. All other tables are shown in the appendix.reference. Seed and model errors are similar in magnitude and as such none will be neglected. In all following figures displaying final results only the combined error is shown.

Comments on difference between these values staying constant? The final results for all selection rules are now visualized. Figure 7 displays final CE estimates using the 5% selection rule. One can observe a general increase in the estimate of conditional entropy for larger bottleneck dimensions, with the exception of the models for $N_B = 8$. The insignificant increase in conditional entropy between $N_B = 25$ and $N_B = 30$ points towards a saturation in the conditional entropy for higher bottleneck dimensions. Figure 8 shows the final CE estimates for the best validation selection rule. This selection rule favors

Table 6: Final CE and MI estimates based on 5% selection rule

N_B	CE [nats] ($\pm\sigma_{\text{seed}} \pm \sigma_{\text{model}}$)	MI [nats] ($\pm\sigma_{\text{seed}} \pm \sigma_{\text{model}}$)	No. samples
4	$36.16\ (\pm 0.49 \pm 0.14)$	$3.27\ (\pm 0.49 \pm 0.10)$	2
6	$36.92\ (\pm 0.32 \pm 0.17)$	$4.02\ (\pm 0.32 \pm 0.13)$	2
8	$39.40\ (\pm 0.16 \pm 0.24)$	$6.51\ (\pm 0.16 \pm 0.17)$	2
15	$38.16\ (\pm 0.39 \pm 0.13)$	$5.27\ (\pm 0.39 \pm 0.09)$	3
16	$38.20\ (\pm 0.35 \pm 0.15)$	$5.31\ (\pm 0.35 \pm 0.11)$	3
25	$38.93\ (\pm 0.34 \pm 0.29)$	$6.04\ (\pm 0.34 \pm 0.22)$	2
30	$39.04\ (\pm - \pm 0.45)$	$6.15\ (\pm - \pm 0.34)$	1

Figure 7: Final CE estimates for x_{HI} inference, 5% rule



models in later stages of training. Models can therefore learn longer and this is the reason why the resulting CE estimates are on average higher. The models have also often begun to overfit, resulting in larger errors than for the previous method. Models with bottleneck dimensions close to the feature distribution are less prone to large errors. Models with bottleneck dimensions farther from that of the feature distribution have learnt less by the time the reach their best validation performance, again with exception of the model for $N_B = 8$. Figure 9 shows the final CE estimates for the last epoch selection rule. The curve and its characteristics are similar to that of the one for the best validation selection rule. It is notable that the model for $N_B = 8$ is estimated to have learned less than the ones for $N_B = 15, 16$, a feature which other selection rules do not have. As this rule chooses models which are stronly overfitting, the model error increases, such that the errors on the final estimates are larger than for the other selection rules.

Small summary here

Figure 8: Final CE estimates for x_{HI} inference, best validation rule

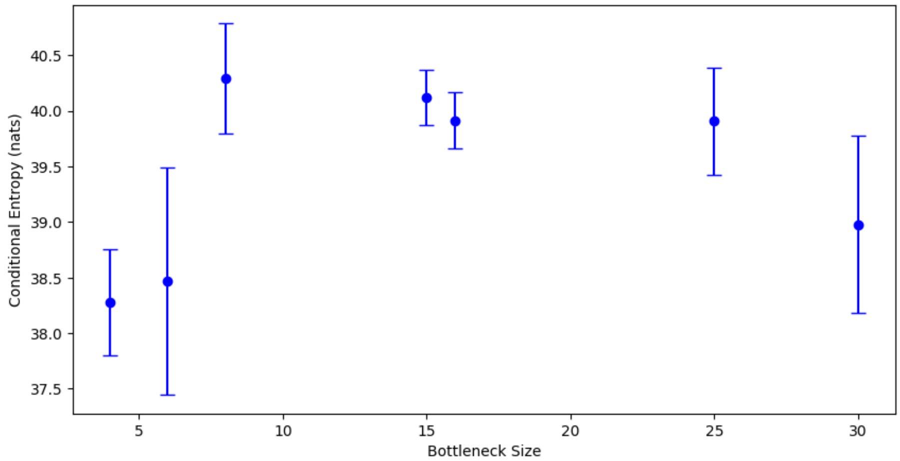
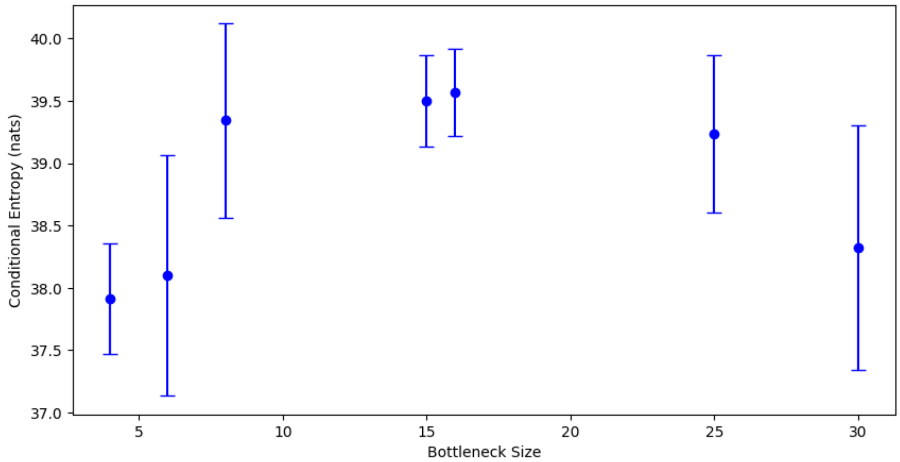


Figure 9: Final CE estimates for x_{HI} inference, last epoch rule



5.2 Estimates for models of the simulation parameters

The 6-parameter inference problem is easier: both model selection rules yield a final MI estimate that monotonously increases with bottleneck dimension N_B. Since the prior distribution is known, the estimate of MI comes with no additional bias towards the CE estimate. As such, in the following only MI estimates will be given.

Training the model

The average training time for this model is insert. Since the CNF is a significantly lighter architecture than the CNN, the CNN pretraining and finetuning stages both have a similar epoch time, while in the CNF pretraining stage, the outputs of the CNN can be reused across epochs. Since the processing of an entire image is much more expensive than that of the CNN output, the pretraining of the flow is accomplished in less than 10 min in all cases. This make this training setup significantly faster than that used for inference of neutral hydrogen.

The bottleneck dimensions analyzed are $N_B \in \{3,4,5,6,7,8,9\}$, each for two different seeds. As an example, the training and validation losses for an ensemble with bottleneck dimension $N_{\text{dim}} = 6$ are shown in figure 10. The loss plots for all other ensembles can be found in the appendix Reference coming soon. The first plot shows the CNN pretraining stage, during which the MSE error drops to around a third its initial value and begins plateauing. The CNF loss quickly drops during its pretraining and has reached a plateau when finetuning begins. Do to the new learning rates used for finetuning, the CNF loss might initially rise at the beginning of finetuning. However in all cases, the finetuning stage does eventually lead to a further improvement in CNF loss. For some N_B, a significant spike in training and validation losses is observed during finetuning, which leads to instability of results achieved with the 5% selection rule.

(a) MSE Losses for CNN pretraining

(b) Flow Losses for CNF pretraining and finetuning

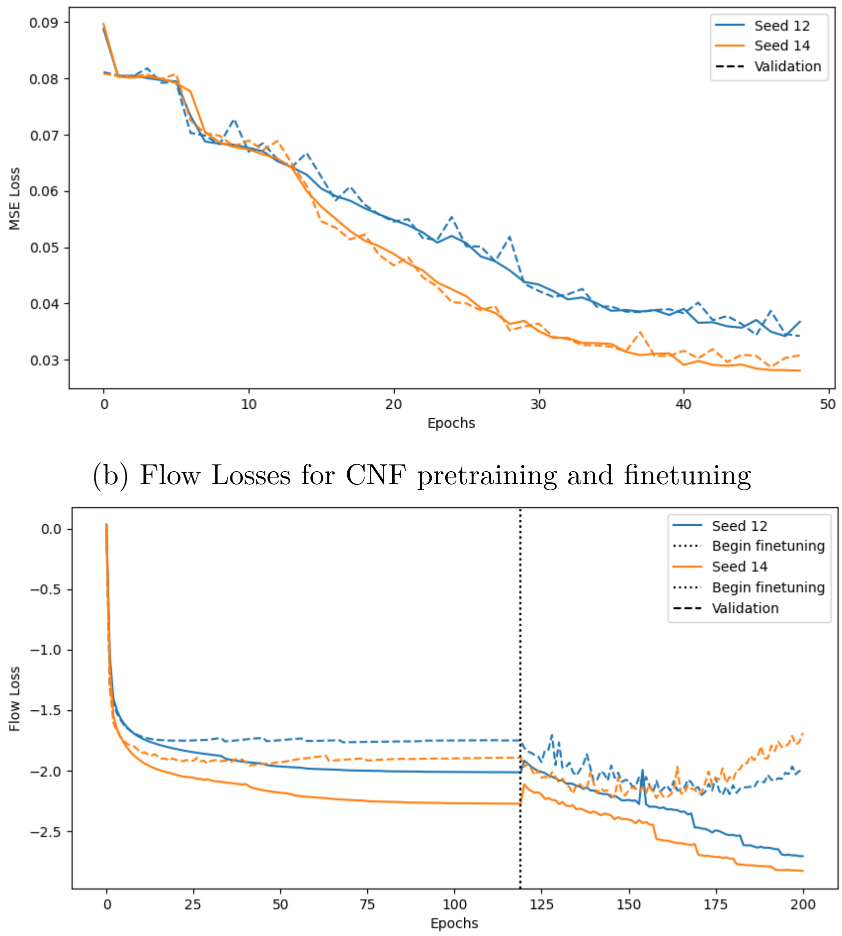
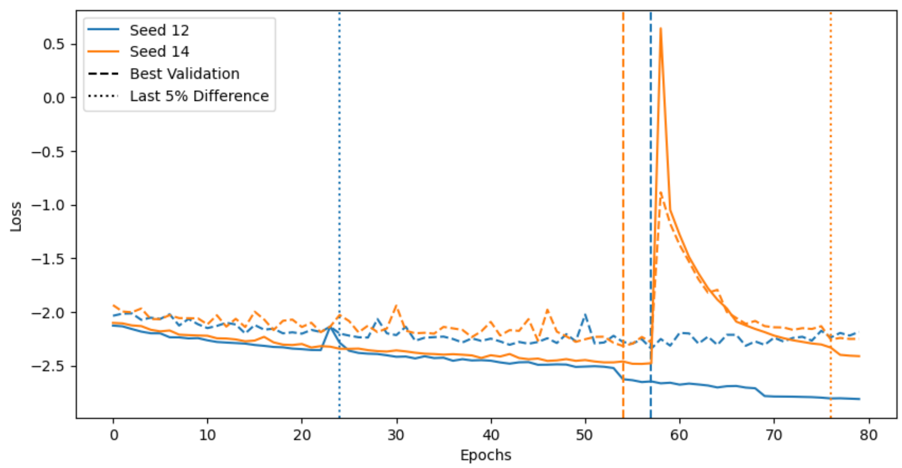


Figure 10: Training and validation losses for $N_B = 6$

To show how the 5% decision rule fails in this case, figure 11 shows training and validation losses for the ensemble $N_B = 7$, where again vertical lines indicate the epoch chosen by best validation rule and dotted lines that chosen by the 5% rule. The model selected by the 5% decision rule leads to a potentially underestimated MI, since earlier models fulfilling a 5% relative distance in training and validation losses performed better in both losses.

Figure 11: Model selection in finetuning for $N_B = 7$



Estimates of conditional entropy and mutual information

The estimates of conditional entropy are obtained in the same way as previously, and since the prior is known, there is no additional bias in estimating mutual information. Hence only MI estimates are shown in this section. In order to compare seed and model errors, table 7 shows

Table 7: Final MI for 6-parameter inference, 5% rule

N_B	MI [nats] ($\pm\sigma_{\text{seed}} \pm \sigma_{\text{model}}$)	No. samples
3	11.66 ± 0.14	2
5	13.12 ± 0.33	2
6	13.33 ± 0.21	2
7	13.93 ± 0.13	2
9	13.92 ± 0.21	2

Models with $N_B = N_{\text{param}}$ seem to have smallest errors and in general behave the most reliably.

6 Interpretation

Still gathering some ideas for the interpretation...

Possible interpretations of the results:

The flow seems to be overcondident, because of the long negative tail in log probability plot. If choosing MI estimation from earlier epochs, get smaller error! This is due to observed overfitting and still improving performance.

The 500 samples are simply not enough to be a precice probe of the distributions being compared. On the other hand, could not have taken more from training set because this would have caused even more training instability. Especially the large seed error points towards this. In total, the 5000 samples seem to be borderline to successfully compare two distributions as high-dimensional as these...

Alternative interpretation:

The CNN learns the latent space. If the weights of the CNN aren't updated enough and the flow is doing too much of the heavy lifting, then the latent space structure might be hard to estimate, resulting in long negative tail.

Check to see wether the extreme models are the reason for this, or if this is a problem with the flow.

The seed uncertainty for this model is very high, given some numbers here. The high sensitivity towards weight initialization points towards unstable training.

One possible explanation for this instability could be the choice of hyperparameters. For example, a large learning rate might magnify an initially chaotic training process, causing the model to end in different minimia of the loss landscape.

Another possibility is the choice of architecture. For example, the flow might be too deep given the data. While one might expect the need for a deep architecture to infer 15-dimensional labels, these labels are likely highly correlated across dimensions because of curve shape etc, making the problem easier than naively expected. The fact that one observes an early saturation in mutual information across bottleneck sizes supports the latter statement.

Of course, there is always the possibility of there being too little training data. Choosing larger parts of the training set for validation and testing indeed showed an increase in overfitting towards the end of training.

Also comment on error changed between CE and MI estiamte

Something about consistency of MI estimate, since e.g. CE estimate is not larger zero from beginning. Maybe cite from paper?

From paper: "In par- ticular, higher estimated MI between observations and learned representations do not seem to indi- cate improved predictive performance when the representations are used for downstream supervised learning tasks (Tschannen et al., 2019)."

A Final estimates of conditional entropy and mutual information

Table 8: Final estimates x_{HI} inference

N_B	CE [nats]	MI [nats]	No. samples
4	$38.28 \pm 0.35 \pm 0.32$	$5.39 \pm 0.35 \pm 0.28$	2
6	$38.47 \pm 0.99 \pm 0.27$	$5.58 \pm 0.99 \pm 0.22$	2
8	$40.29 \pm 0.36 \pm 0.34$	$7.40 \pm 0.36 \pm 0.25$	2
15	$40.12 \pm 0.16 \pm 0.18$	$7.23 \pm 0.16 \pm 0.14$	3
16	$39.91 \pm 0.10 \pm 0.23$	$7.02 \pm 0.10 \pm 0.19$	3
25	$39.91 \pm 0.29 \pm 0.38$	$7.02 \pm 0.29 \pm 0.31$	2
30	$38.98 \pm - \pm 0.80$	$6.09 \pm - \pm 0.69$	1

Table 9: Final estimates x_{HI} inference under constraints

Bottleneck	CE [nats]	MI [nats]	No. samples
4	36.16 ± 0.51	3.27 ± 0.50	2
6	36.92 ± 0.36	4.02 ± 0.34	2
8	39.40 ± 0.28	6.51 ± 0.23	2
15	38.16 ± 0.41	5.27 ± 0.40	3
16	38.20 ± 0.38	5.31 ± 0.37	3
25	38.93 ± 0.45	6.04 ± 0.41	2
30	39.04 ± 0.45	6.15 ± 0.34	1

Table 10: Final estimates x_{HI} inference on last-epoch models

Bottleneck	CE [nats]	MI [nats]	No. samples
4	37.92 ± 0.44	5.02 ± 0.40	2
6	38.10 ± 0.97	5.21 ± 0.95	2
8	39.34 ± 0.78	6.45 ± 0.73	2
15	39.50 ± 0.37	6.61 ± 0.34	3
16	39.57 ± 0.35	6.68 ± 0.31	3
25	39.24 ± 0.63	6.34 ± 0.57	2
30	38.33 ± 0.98	5.44 ± 0.87	1

Table 11: CE estimates across bottleneck dims N_B in nats

N_{dim} \ Seed	12	14	30
4	38.88 ± 0.89	-	37.68 ± 0.42
6	36.75 ± 0.40	-	40.25 ± 0.70
15	39.64 ± 0.52	40.15 ± 0.59	40.61 ± 0.37
16	39.81 ± 0.38	40.17 ± 0.50	39.72 ± 0.68
25	-	-	40.22 ± 0.57

Table 12: MI estimates across bottleneck dims N_{dim} in nats

N_{dim} \ Seed	12	14	30
4	6.00 ± 0.73	-	4.76 ± 0.31
6	3.87 ± 0.30	-	7.24 ± 0.61
15	6.70 ± 0.45	7.25 ± 0.48	7.74 ± 0.25
16	6.94 ± 0.29	7.31 ± 0.39	6.81 ± 0.57
25	-	-	7.34 ± 0.47

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Acknowledgements

Erklärung

Ich versichere, dass ich diese Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe. Ich erkläre, dass ich beim Schreiben des in der Arbeit verwendeten Programmcodes den Coding-Agent Copilot Pro benutzt habe. Copilot Pro hilft vor allem bei der Strukturierung und mit der Fehlerfindung im Code. Alle wissenschaftlichen Inhalte, Analysen und Schlussfolgerungen wurden von mir selbst erarbeitet und verantwortet.

Heidelberg, den 13.05.2026,